

Ryan Cho

Staff Software Engineer

+1 608 360 1459 · Ryanmcho100@gmail.com · <https://www.linkedin.com/in/ryan-cho-2020/> ·
<https://github.com/Chomama> · New York, United States

PROFESSIONAL SUMMARY

Full-stack software engineer with 7+ years shipping production web applications across finance and SaaS. Deep expertise in Angular and TypeScript frontend architecture, Python/Django backend services, and secure enterprise API integrations with JWT and OAuth2. Track record building real-time UI modules, microservices, and PostgreSQL-backed platforms while using AI-assisted development to maintain high delivery velocity and code quality.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Owned **end-to-end feature delivery** from PostgreSQL schema design through Angular/TypeScript UI and Python/Django APIs for a capital markets platform supporting **500+ daily active users** across portfolio and risk workflows.
- Established a **modular Angular architecture** organized by product capabilities with lazy-loaded routes and shared component libraries, reducing initial bundle size by **35%** and shortening time-to-ship for new trading screens.
- Engineered **HTTP interceptor chains** for token refresh, structured error capture, and retry handling across the API layer, improving reliability on authentication-heavy workflows and standardizing client-side failure recovery.
- Delivered **real-time monitoring modules** using WebSockets, Server-Sent Events, and RxJS state orchestration to stream position and exposure data with sub-second refresh across concurrent browser sessions.
- Built **Python microservices** for trade validation, settlement processing, and audit logging with JWT service-to-service auth and role-scoped permission enforcement across multi-tenant account boundaries.
- Reduced **p95 API latency from 420ms to 95ms** by optimizing PostgreSQL indexes, query plans, and ORM access patterns on high-frequency reporting endpoints used by operations and compliance teams.
- Implemented **localization workflows and PWA caching** with service workers, enabling offline portfolio snapshots and consistent formatting for US and European client configurations.
- Hardened **OAuth2 and OpenID Connect authorization** flows for internal portals and automated integrations, ensuring human users and background jobs operated within strict permission scopes.
- Integrated **AI coding agents** into planning, scaffolding, and peer-review workflows, accelerating delivery of tested UI modules and API endpoints while preserving team quality gates before production release.

Technology Analyst, Incoming Technology Analyst, Technology

May/2019 - Apr/2022

Summer Analyst,

BNP Paribas

Remote | Seventh Avenue, New York

- Shipped **production-grade commercial web applications** for global markets operations, supporting fixed-income and derivatives workflows across three business lines under strict regulatory audit requirements.
- Developed **Angular and React client applications** connected to RESTful microservices, translating complex trade and reference data into resilient, form-driven experiences for desk analysts and operations staff.
- Extended **Python and Java backend services** with routing logic, validation layers, and batch reconciliation jobs consumed by front-office and middle-office systems across the trading lifecycle.
- Implemented **OAuth2 and JWT authentication** with centralized token validation and session management, ensuring secure access boundaries for internal portals and automated downstream consumers.
- Designed **request/response interceptors and error middleware** that standardized telemetry capture, retry behavior, and user-facing recovery states across multiple client codebases.
- Improved **PostgreSQL and MS SQL performance** on position and reporting queries by refactoring stored procedures and ORM access paths, increasing throughput for high-volume trade batch processing.
- Contributed to **microservice extraction** from monolithic reporting modules into independently deployable services with contract-tested API boundaries and automated CI/CD deployment pipelines.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Built **React and TypeScript UI components** with REST API integrations for a food-safety compliance platform, enabling restaurant operators to manage inventory and inspection records from a unified dashboard.
- Developed **Django backend endpoints** for user onboarding, document uploads, and notification triggers that automated compliance workflows for early enterprise pilot customers.
- Modeled **PostgreSQL schemas and migrations** for multi-location account hierarchies, maintaining referential integrity as pilot deployments expanded across restaurant groups.
- Collaborated with product and design to deliver **responsive form modules** with client-side validation, reducing submission errors during high-traffic onboarding periods.
- Contributed tested feature branches through **code reviews and sprint planning**, producing documentation that accelerated handoff to full-time engineering hires.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Developed **responsive public-facing web pages** for a nonprofit foundation using accessible HTML, CSS, and JavaScript layouts with cross-browser compatibility for campaign launches.
- Integrated **third-party donation and newsletter APIs** to enable secure form submissions and subscriber capture without exposing credentials on the client.
- Improved **mobile page load performance** through asset minification and image optimization, meeting Lighthouse targets ahead of fundraising event deadlines.
- Maintained **CMS-driven content templates** so non-technical stakeholders could independently publish event and grant program updates.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: Angular, TypeScript, React.js, AngularJS, RxJS, Angular Material, Bootstrap, HTML5, CSS3, PWA, Service Workers, i18n/Localization, State Management

Backend: Python, Django, FastAPI, REST APIs, Microservices, Node.js, JWT, OAuth2, OpenID Connect, WebSockets, SSE

Databases: PostgreSQL, MS SQL, SQL, Query Optimization, Schema Design, Indexing

Cloud & DevOps: Docker, CI/CD, AWS, Git, Linux, Containerized Deployments

Monitoring & Tools: Application Performance Monitoring, Structured Logging, Postman, Chrome DevTools, ESLint, Prettier

Workflow & PM: Agile/Scrum, Jira, Code Review, Technical Mentorship, Cross-Team Collaboration, AI-Assisted Development